

Example 2.51. Let X be a Moore space $M(\mathbb{Z}_m, n)$ obtained from S^n by attaching a cell e^{n+1} by a map of degree m . The quotient map $f: X \rightarrow X/S^n = S^{n+1}$ induces trivial homomorphisms on reduced homology with $\mathbb{Z}$ coefficients since the nonzero reduced homology groups of X and S^{n+1} occur in different dimensions. But with $\mathbb{Z}_m$ coefficients the story is different, as we can see by considering the long exact sequence of the pair (X, S^n) , which contains the segment

$$0 = \tilde{H}_{n+1}(S^n; \mathbb{Z}_m) \rightarrow \tilde{H}_{n+1}(X; \mathbb{Z}_m) \xrightarrow{f_*} \tilde{H}_{n+1}(X/S^n; \mathbb{Z}_m)$$

Exactness says that f_* is injective, hence nonzero since $\tilde{H}_{n+1}(X; \mathbb{Z}_m)$ is $\mathbb{Z}_m$, the cellular boundary map $H_{n+1}(X^{n+1}, X^n; \mathbb{Z}_m) \rightarrow H_n(X^n, X^{n-1}; \mathbb{Z}_m)$ being $\mathbb{Z}_m \xrightarrow{m} \mathbb{Z}_m$.

Exercises

1. Prove the Brouwer fixed point theorem for maps $f: D^n \rightarrow D^n$ by applying degree theory to the map $S^n \rightarrow S^n$ that sends both the northern and southern hemispheres of S^n to the southern hemisphere via f . [This was Brouwer's original proof.]
2. Given a map $f: S^{2n} \rightarrow S^{2n}$, show that there is some point $x \in S^{2n}$ with either $f(x) = x$ or $f(x) = -x$. Deduce that every map $\mathbb{R}P^{2n} \rightarrow \mathbb{R}P^{2n}$ has a fixed point. Construct maps $\mathbb{R}P^{2n-1} \rightarrow \mathbb{R}P^{2n-1}$ without fixed points from linear transformations $\mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ without eigenvectors.
3. Let $f: S^n \rightarrow S^n$ be a map of degree zero. Show that there exist points $x, y \in S^n$ with $f(x) = x$ and $f(y) = -y$. Use this to show that if F is a continuous vector field defined on the unit ball D^n in $\mathbb{R}^n$ such that $F(x) \neq 0$ for all x , then there exists a point on ∂D^n where F points radially outward and another point on ∂D^n where F points radially inward.
4. Construct a surjective map $S^n \rightarrow S^n$ of degree zero, for each $n \geq 1$.
5. Show that any two reflections of S^n across different n -dimensional hyperplanes are homotopic, in fact homotopic through reflections. [The linear algebra formula for a reflection in terms of inner products may be helpful.]
6. Show that every map $S^n \rightarrow S^n$ can be homotoped to have a fixed point if $n > 0$.
7. For an invertible linear transformation $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ show that the induced map on $H_n(\mathbb{R}^n, \mathbb{R}^n - \{0\}) \approx \tilde{H}_{n-1}(\mathbb{R}^n - \{0\}) \approx \mathbb{Z}$ is $\mathbb{1}$ or $-\mathbb{1}$ according to whether the determinant of f is positive or negative. [Use Gaussian elimination to show that the matrix of f can be joined by a path of invertible matrices to a diagonal matrix with ± 1 's on the diagonal.]
8. A polynomial $f(z)$ with complex coefficients, viewed as a map $\mathbb{C} \rightarrow \mathbb{C}$, can always be extended to a continuous map of one-point compactifications $\hat{f}: S^2 \rightarrow S^2$. Show that the degree of $\hat{f}$ equals the degree of f as a polynomial. Show also that the local degree of $\hat{f}$ at a root of f is the multiplicity of the root.

9. Compute the homology groups of the following 2-complexes:
- (a) The quotient of S^2 obtained by identifying north and south poles to a point.
 - (b) $S^1 \times (S^1 \vee S^1)$.
 - (c) The space obtained from D^2 by first deleting the interiors of two disjoint subdisks in the interior of D^2 and then identifying all three resulting boundary circles together via homeomorphisms preserving clockwise orientations of these circles.
 - (d) The quotient space of $S^1 \times S^1$ obtained by identifying points in the circle $S^1 \times \{x_0\}$ that differ by $2\pi/m$ rotation and identifying points in the circle $\{x_0\} \times S^1$ that differ by $2\pi/n$ rotation.
10. Let X be the quotient space of S^2 under the identifications $x \sim -x$ for x in the equator S^1 . Compute the homology groups $H_i(X)$. Do the same for S^3 with antipodal points of the equatorial $S^2 \subset S^3$ identified.
11. In an exercise for §1.2 we described a 3-dimensional CW complex obtained from the cube I^3 by identifying opposite faces via a one-quarter twist. Compute the homology groups of this complex.
12. Show that the quotient map $S^1 \times S^1 \rightarrow S^2$ collapsing the subspace $S^1 \vee S^1$ to a point is not nullhomotopic by showing that it induces an isomorphism on H_2 . On the other hand, show via covering spaces that any map $S^2 \rightarrow S^1 \times S^1$ is nullhomotopic.
13. Let X be the 2-complex obtained from S^1 with its usual cell structure by attaching two 2-cells by maps of degrees 2 and 3, respectively.
- (a) Compute the homology groups of all the subcomplexes $A \subset X$ and the corresponding quotient complexes X/A .
 - (b) Show that $X \simeq S^2$ and that the only subcomplex $A \subset X$ for which the quotient map $X \rightarrow X/A$ is a homotopy equivalence is the trivial subcomplex, the 0-cell.
14. A map $f: S^n \rightarrow S^n$ satisfying $f(x) = f(-x)$ for all x is called an *even map*. Show that an even map $S^n \rightarrow S^n$ must have even degree, and that the degree must in fact be zero when n is even. When n is odd, show there exist even maps of any given even degree. [Hints: If f is even, it factors as a composition $S^n \rightarrow \mathbb{RP}^n \rightarrow S^n$. Using the calculation of $H_n(\mathbb{RP}^n)$ in the text, show that the induced map $H_n(S^n) \rightarrow H_n(\mathbb{RP}^n)$ sends a generator to twice a generator when n is odd. It may be helpful to show that the quotient map $\mathbb{RP}^n \rightarrow \mathbb{RP}^n/\mathbb{RP}^{n-1}$ induces an isomorphism on H_n when n is odd.]
15. Show that if X is a CW complex then $H_n(X^n)$ is free by identifying it with the kernel of the cellular boundary map $H_n(X^n, X^{n-1}) \rightarrow H_{n-1}(X^{n-1}, X^{n-2})$.
16. Let $\Delta^n = [v_0, \dots, v_n]$ have its natural Δ -complex structure with k -simplices $[v_{i_0}, \dots, v_{i_k}]$ for $i_0 < \dots < i_k$. Compute the ranks of the simplicial (or cellular) chain groups $\Delta_i(\Delta^n)$ and the subgroups of cycles and boundaries. [Hint: Pascal's triangle.] Apply this to show that the k -skeleton of Δ^n has homology groups $\tilde{H}_i((\Delta^n)^k)$ equal to 0 for $i < k$, and free of rank $\binom{n}{k+1}$ for $i = k$.

17. Show the isomorphism between cellular and singular homology is natural in the following sense: A map $f: X \rightarrow Y$ that is *cellular* — satisfying $f(X^n) \subset Y^n$ for all n — induces a chain map f_* between the cellular chain complexes of X and Y , and the map $f_*: H_n^{CW}(X) \rightarrow H_n^{CW}(Y)$ induced by this chain map corresponds to $f_*: H_n(X) \rightarrow H_n(Y)$ under the isomorphism $H_n^{CW} \approx H_n$.
18. For a CW pair (X, A) show there is a relative cellular chain complex formed by the groups $H_i(X^i, X^{i-1} \cup A^i)$, having homology groups isomorphic to $H_n(X, A)$.
19. Compute $H_i(\mathbb{R}P^n / \mathbb{R}P^m)$ for $m < n$ by cellular homology, using the standard CW structure on $\mathbb{R}P^n$ with $\mathbb{R}P^m$ as its m -skeleton.
20. For finite CW complexes X and Y , show that $\chi(X \times Y) = \chi(X)\chi(Y)$.
21. If a finite CW complex X is the union of subcomplexes A and B , show that $\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B)$.
22. For X a finite CW complex and $p: \tilde{X} \rightarrow X$ an n -sheeted covering space, show that $\chi(\tilde{X}) = n\chi(X)$.
23. Show that if the closed orientable surface M_g of genus g is a covering space of M_h , then $g = n(h-1) + 1$ for some n , namely, n is the number of sheets in the covering. [Conversely, if $g = n(h-1) + 1$ then there is an n -sheeted covering $M_g \rightarrow M_h$, as we saw in Example 1.41.]
24. Suppose we build S^2 from a finite collection of polygons by identifying edges in pairs. Show that in the resulting CW structure on S^2 the 1-skeleton cannot be either of the two graphs shown, with five and six vertices. [This is one step in a proof that neither of these graphs embeds in $\mathbb{R}^2$.]
- The image shows two graphs side-by-side. The left graph has 5 vertices arranged in a regular pentagon. Every pair of vertices is connected by an edge, forming a complete graph K_5 . The right graph has 6 vertices arranged in a regular hexagon. Every pair of vertices is connected by an edge, forming a complete graph K_6 .
25. Show that for each $n \in \mathbb{Z}$ there is a unique function φ assigning an integer to each finite CW complex, such that (a) $\varphi(X) = \varphi(Y)$ if X and Y are homeomorphic, (b) $\varphi(X) = \varphi(A) + \varphi(X/A)$ if A is a subcomplex of X , and (c) $\varphi(S^0) = n$. For such a function φ , show that $\varphi(X) = \varphi(Y)$ if $X \simeq Y$.
26. For a pair (X, A) , let $X \cup CA$ be X with a cone on A attached.
- (a) Show that X is a retract of $X \cup CA$ iff A is *contractible in X* : There is a homotopy $f_t: A \rightarrow X$ with f_0 the inclusion $A \hookrightarrow X$ and f_1 a constant map.
- (b) Show that if A is contractible in X then $H_n(X, A) \approx \tilde{H}_n(X) \oplus \tilde{H}_{n-1}(A)$, using the fact that $(X \cup CA)/X$ is the suspension SA of A .
27. The short exact sequences $0 \rightarrow C_n(A) \rightarrow C_n(X) \rightarrow C_n(X, A) \rightarrow 0$ always split, but why does this not always yield splittings $H_n(X) \approx H_n(A) \oplus H_n(X, A)$?
28. (a) Use the Mayer-Vietoris sequence to compute the homology groups of the space obtained from a torus $S^1 \times S^1$ by attaching a Möbius band via a homeomorphism from the boundary circle of the Möbius band to the circle $S^1 \times \{x_0\}$ in the torus.
- (b) Do the same for the space obtained by attaching a Möbius band to $\mathbb{R}P^2$ via a homeomorphism of its boundary circle to the standard $\mathbb{R}P^1 \subset \mathbb{R}P^2$.

29. The surface M_g of genus g , embedded in $\mathbb{R}^3$ in the standard way, bounds a compact region R . Two copies of R , glued together by the identity map between their boundary surfaces M_g , form a closed 3-manifold X . Compute the homology groups of X via the Mayer-Vietoris sequence for this decomposition of X into two copies of R . Also compute the relative groups $H_i(R, M_g)$.
30. For the mapping torus T_f of a map $f: X \rightarrow X$, we constructed in Example 2.48 a long exact sequence $\cdots \rightarrow H_n(X) \xrightarrow{1-f_*} H_n(X) \rightarrow H_n(T_f) \rightarrow H_{n-1}(X) \rightarrow \cdots$. Use this to compute the homology of the mapping tori of the following maps:
- (a) A reflection $S^2 \rightarrow S^2$.
 - (b) A map $S^2 \rightarrow S^2$ of degree 2.
 - (c) The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that is the identity on one factor and a reflection on the other.
 - (d) The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that is a reflection on each factor.
 - (e) The map $S^1 \times S^1 \rightarrow S^1 \times S^1$ that interchanges the two factors and then reflects one of the factors.
31. Use the Mayer-Vietoris sequence to show there are isomorphisms $\tilde{H}_n(X \vee Y) \approx \tilde{H}_n(X) \oplus \tilde{H}_n(Y)$ if the basepoints of X and Y that are identified in $X \vee Y$ are deformation retracts of neighborhoods $U \subset X$ and $V \subset Y$.
32. For SX the suspension of X , show by a Mayer-Vietoris sequence that there are isomorphisms $\tilde{H}_n(SX) \approx \tilde{H}_{n-1}(X)$ for all n .
33. Suppose the space X is the union of open sets $A_1, \dots, A_n$ such that each intersection $A_{i_1} \cap \cdots \cap A_{i_k}$ is either empty or has trivial reduced homology groups. Show that $\tilde{H}_i(X) = 0$ for $i \geq n-1$, and give an example showing this inequality is best possible, for each n .
34. Derive the long exact sequence of a pair (X, A) from the Mayer-Vietoris sequence applied to $X \cup CA$, where CA is the cone on A . [We showed after the proof of Proposition 2.22 that $H_n(X, A) \approx \tilde{H}_n(X \cup CA)$ for all n .]
35. Use the Mayer-Vietoris sequence to show that a nonorientable closed surface, or more generally a finite simplicial complex X for which $H_1(X)$ contains torsion, cannot be embedded as a subspace of $\mathbb{R}^3$ in such a way as to have a neighborhood homeomorphic to the mapping cylinder of some map from a closed orientable surface to X . [This assumption on a neighborhood is in fact not needed if one deduces the result from Alexander duality in §3.3.]
36. Show that $H_i(X \times S^n) \approx H_i(X) \oplus H_{i-n}(X)$ for all i and n , where $H_i = 0$ for $i < 0$ by definition. Namely, show $H_i(X \times S^n) \approx H_i(X) \oplus H_i(X \times S^n, X \times \{x_0\})$ and $H_i(X \times S^n, X \times \{x_0\}) \approx H_{i-1}(X \times S^{n-1}, X \times \{x_0\})$. [For the latter isomorphism the relative Mayer-Vietoris sequence yields an easy proof.]
37. Give an elementary derivation for the Mayer-Vietoris sequence in simplicial homology for a Δ -complex X decomposed as the union of subcomplexes A and B .

38. Show that a commutative diagram

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & C_{n+1} & \searrow & B_n & \longrightarrow & C_n \\
 & & \downarrow & & \downarrow & & \downarrow \\
 \cdots & \longrightarrow & E_{n+1} & \searrow & D_n & \longrightarrow & E_n \\
 & & & & & & \searrow \\
 & & & & & & A_{n-1} \\
 & & & & & & \searrow \\
 & & & & & & B_{n-1} \longrightarrow \cdots \\
 & & & & & & \downarrow \\
 & & & & & & D_{n-1} \longrightarrow \cdots
 \end{array}$$

with the two sequences across the top and bottom exact, gives rise to an exact sequence $\cdots \rightarrow E_{n+1} \rightarrow B_n \rightarrow C_n \oplus D_n \rightarrow E_n \rightarrow B_{n-1} \rightarrow \cdots$ where the maps are obtained from those in the previous diagram in the obvious way, except that $B_n \rightarrow C_n \oplus D_n$ has a minus sign in one coordinate.

39. Use the preceding exercise to derive relative Mayer-Vietoris sequences for CW pairs $(X, Y) = (A \cup B, C \cup D)$ with $A = B$ or $C = D$.

40. From the long exact sequence of homology groups associated to the short exact sequence of chain complexes $0 \rightarrow C_i(X) \xrightarrow{n} C_i(X) \rightarrow C_i(X; \mathbb{Z}_n) \rightarrow 0$ deduce immediately that there are short exact sequences

$$0 \rightarrow H_i(X)/nH_i(X) \rightarrow H_i(X; \mathbb{Z}_n) \rightarrow n\text{-Torsion}(H_{i-1}(X)) \rightarrow 0$$

where $n\text{-Torsion}(G)$ is the kernel of the map $G \xrightarrow{n} G$, $g \mapsto ng$. Use this to show that $\tilde{H}_i(X; \mathbb{Z}_p) = 0$ for all i and all primes p iff $\tilde{H}_i(X)$ is a vector space over $\mathbb{Q}$ for all i .

41. For X a finite CW complex and F a field, show that the Euler characteristic $\chi(X)$ can also be computed by the formula $\chi(X) = \sum_n (-1)^n \dim H_n(X; F)$, the alternating sum of the dimensions of the vector spaces $H_n(X; F)$.

42. Let X be a finite connected graph having no vertex that is the endpoint of just one edge, and suppose that $H_1(X; \mathbb{Z})$ is free abelian of rank $n > 1$, so the group of automorphisms of $H_1(X; \mathbb{Z})$ is $GL_n(\mathbb{Z})$, the group of invertible $n \times n$ matrices with integer entries whose inverse matrix also has integer entries. Show that if G is a finite group of homeomorphisms of X , then the homomorphism $G \rightarrow GL_n(\mathbb{Z})$ assigning to $g: X \rightarrow X$ the induced homomorphism $g_*: H_1(X; \mathbb{Z}) \rightarrow H_1(X; \mathbb{Z})$ is injective. Show the same result holds if the coefficient group $\mathbb{Z}$ is replaced by $\mathbb{Z}_m$ with $m > 2$. What goes wrong when $m = 2$?

43. (a) Show that a chain complex of free abelian groups C_n splits as a direct sum of subcomplexes $0 \rightarrow L_{n+1} \rightarrow K_n \rightarrow 0$ with at most two nonzero terms. [Show the short exact sequence $0 \rightarrow \text{Ker } \partial \rightarrow C_n \rightarrow \text{Im } \partial \rightarrow 0$ splits and take $K_n = \text{Ker } \partial$.]

(b) In case the groups C_n are finitely generated, show there is a further splitting into summands $0 \rightarrow \mathbb{Z} \rightarrow 0$ and $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow 0$. [Reduce the matrix of the boundary map $L_{n+1} \rightarrow K_n$ to echelon form by elementary row and column operations.]

(c) Deduce that if X is a CW complex with finitely many cells in each dimension, then $H_n(X; G)$ is the direct sum of the following groups:

- a copy of G for each $\mathbb{Z}$ summand of $H_n(X)$
- a copy of G/mG for each $\mathbb{Z}_m$ summand of $H_n(X)$
- a copy of the kernel of $G \xrightarrow{m} G$ for each $\mathbb{Z}_m$ summand of $H_{n-1}(X)$

2.3 The Formal Viewpoint

Sometimes it is good to step back from the forest of details and look for general patterns. In this rather brief section we will first describe the general pattern of homology by axioms, then we will look at some common formal features shared by many of the constructions we have made, using the language of categories and functors which has become common in much of modern mathematics.

Axioms for Homology

For simplicity let us restrict attention to CW complexes and focus on reduced homology to avoid mentioning relative homology. A (reduced) **homology theory** assigns to each nonempty CW complex X a sequence of abelian groups $\tilde{h}_n(X)$ and to each map $f: X \rightarrow Y$ between CW complexes a sequence of homomorphisms $f_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$ such that $(fg)_* = f_*g_*$ and $\mathbb{1}_* = \mathbb{1}$, and so that the following three axioms are satisfied.

- (1) If $f \simeq g: X \rightarrow Y$, then $f_* = g_*: \tilde{h}_n(X) \rightarrow \tilde{h}_n(Y)$.
- (2) There are boundary homomorphisms $\partial: \tilde{h}_n(X/A) \rightarrow \tilde{h}_{n-1}(A)$ defined for each CW pair (X, A) , fitting into an exact sequence

$$\cdots \xrightarrow{\partial} \tilde{h}_n(A) \xrightarrow{i_*} \tilde{h}_n(X) \xrightarrow{q_*} \tilde{h}_n(X/A) \xrightarrow{\partial} \tilde{h}_{n-1}(A) \xrightarrow{i_*} \cdots$$

where i is the inclusion and q is the quotient map. Furthermore the boundary maps are natural: For $f: (X, A) \rightarrow (Y, B)$ inducing a quotient map $\bar{f}: X/A \rightarrow Y/B$, there are commutative diagrams

$$\begin{array}{ccc} \tilde{h}_n(X/A) & \xrightarrow{\partial} & \tilde{h}_{n-1}(A) \\ \downarrow \bar{f}_* & & \downarrow f_* \\ \tilde{h}_n(Y/B) & \xrightarrow{\partial} & \tilde{h}_{n-1}(B) \end{array}$$

- (3) For a wedge sum $X = \bigvee_{\alpha} X_{\alpha}$ with inclusions $i_{\alpha}: X_{\alpha} \hookrightarrow X$, the direct sum map $\bigoplus_{\alpha} i_{\alpha*}: \bigoplus_{\alpha} \tilde{h}_n(X_{\alpha}) \rightarrow \tilde{h}_n(X)$ is an isomorphism for each n .

Negative values for the subscripts n are permitted. Ordinary singular homology is zero in negative dimensions by definition, but interesting homology theories with nontrivial groups in negative dimensions do exist.

The third axiom may seem less substantial than the first two, and indeed for finite wedge sums it can be deduced from the first two axioms, though not in general for infinite wedge sums, as an example in the Exercises shows.

It is also possible, and not much more difficult, to give axioms for unreduced homology theories. One supposes one has relative groups $h_n(X, A)$ defined, specializing to absolute groups by setting $h_n(X) = h_n(X, \emptyset)$. Axiom (1) is replaced by its